

Feedback and Stability

Course notes - Department of Electrical Engineering

1. Open and closed loop

An open-loop controller applies a predetermined input and cannot correct for disturbance or model error. A closed-loop controller measures the output and drives the error to zero.

Feedback buys robustness to modelling error and disturbance rejection. It costs stability margin, which is the central trade of the subject.

2. Transfer functions

Applying the Laplace transform to a linear time-invariant differential equation turns it into an algebraic ratio of polynomials, the transfer function. Poles are the roots of the denominator and determine the natural response; zeros shape how strongly modes are excited.

A second-order system is characterised by its natural frequency and damping ratio, which set overshoot, rise time and settling time.

3. Stability

A continuous linear system is stable when every pole lies strictly in the left half-plane. The Routh-Hurwitz criterion determines this from the coefficients without computing the roots.

For discrete systems the corresponding region is the interior of the unit circle. Marginal stability, with poles exactly on the boundary, is a mathematical case rather than a practical design target.

4. Root locus and frequency response

The root locus traces closed-loop pole positions as a gain varies, showing directly how much gain the loop tolerates before instability.

Bode plots show magnitude and phase against frequency. Gain margin and phase margin quantify how much extra gain or delay the loop can absorb; a phase margin near 45 to 60 degrees is a common design target. The Nyquist criterion relates encirclements of the critical point to closed-loop stability and handles cases Bode analysis cannot.

5. PID control

Proportional action responds to present error, integral action eliminates steady-state error by accumulating history, and derivative action anticipates by responding to the rate of change.

Integral windup occurs when the actuator saturates while the integral term keeps accumulating, producing large overshoot on recovery; clamping the integrator prevents it. Derivative action amplifies measurement noise and is normally filtered or omitted.